AD

ARGUS

# ARGUS DRIVE

## Serial Command Interface Specification

USB Serial (UART0) · 115200 Baud · 8N1

**FIRMWARE v4.1.3**

DOCUMENT REVISION 1.0

Command Reference · Actions · Examples · Safety Restrictions

# 1. Overview

ARGUS DRIVE exposes a binary/hex serial command interface over the USB CDC port (UART0). Every command is a single command byte followed by optional argument bytes, terminated with a newline character (**\n**, 0x0A). Responses are returned on the same port.

| Parameter | Value | Notes |
|---|---|---|
| `Port` | USB Serial (UART0) | CDC / virtual COM port |
| `Baud rate` | 115200 | Fixed; not configurable at runtime |
| `Frame format` | 8N1 | 8 data bits, no parity, 1 stop bit |
| `Terminator` | LF (\n, 0x0A) | Append to every command sent |
| `Firmware` | v4.1.3 | Reported by command 0x01 (INFO) |

**Sending a command**
Transmit the command byte, then any argument bytes, then a newline (`\n`). Example: to steer right send `0x05 0x01 0x0A`. Argument values shown as 00-64 are hexadecimal (0-100 decimal).

## Frame format

Example - steer right: **`0x05 01 0A`**. Commands without arguments are a single byte plus the terminator.

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## Command groups

| | | | |
|---|---|---|---|
| | **Feedback** | `0x01-0x04` | Reads and acceleration feedback |
| | **Steer/Brake** | `0x05/06, 0x12-14` | Actuator control, normal and admin |
| | **Vehicle** | `0x07-0x0B` | Speed, gear, lights, indicators, horn |
| | **System** | `0x0C-11, 0x16` | Reset, Wi-Fi, limits, full status read |

# 2. Command List

| Cmd | Name | Action | Arguments | Group |
|---|---|---|---|---|
| 0x01 | INFO | Get firmware version | None | System |
| 0x02 | ACCEL | Set acceleration (PWM) | 00-64 (0-100%) | Feedback |
| 0x03 | BRAKE | Read brake feedback | None | Feedback |
| 0x04 | STEER | Read steering angle | None | Feedback |
| 0x05 | STR_SW | Control steering actuator | 00 Stop / 01 Left / 02 Right | Steer/Brake |
| 0x06 | BRK_SW | Control brake actuator | 00 Stop / 01 Fwd / 02 Rev | Steer/Brake |
| 0x07 | SPEED | Set speed mode | 01 / 02 / 03 | Vehicle |
| 0x08 | GEAR | Set gear | 01 Drive / 02 Neutral / 03 Reverse | Vehicle |
| 0x09 | LIGHT | Control lights | 00 Off / 01 Ring / 02 High / 03 Low | Vehicle |
| 0x0A | IND | Control indicators | 00 Off / 01 Left / 02 Right / 03 Parking | Vehicle |
| 0x0B | HORN | Activate horn | 01 0.5s / 02 1s / 03 2s | Vehicle |
| 0x0C | RESET | Restart module | None | System |
| 0x0D | WIFI_S | Set Wi-Fi credentials | <SSID> <Password> | System |
| 0x0E | WIFI | Set Wi-Fi mode | 00 Off / 01 STA / 02 AP | System |
| 0x0F | BRK_LIM | Set/read brake angle limits | <Min> <Max> | System |
| 0x10 | STR_LIM | Set/read steering angle limits | <Min> <Max> | System |
| 0x11 | LIM_EN | Enable/disable steering limits | 00 Disable / 01 Enable | System |
| 0x12 | STR_ADM | Admin steering control | 00 Stop / 01 Left / 02 Right | Steer/Brake |
| 0x13 | BRK_ADM | Admin brake control | 00 Stop / 01 Fwd / 02 Rev | Steer/Brake |
| 0x14 | STR_CTR | Automatically centre steering | None | Steer/Brake |
| 0x16 | STATUS | Read all switch states | None | System |

0x15 is not assigned. Locked commands (see Section 5) are documented here but rejected while locked.

# 3. Command Actions

| Acceleration and feedback | |
|---|---|
| 0x01 | Get firmware version. |
| 0x02 | Set acceleration level. Actual duty is clamped to 20-90%, except when braking. |
| 0x03 | Read brake feedback in 0.1% units. |
| 0x04 | Read steering angle in 0.1° units. |

| Steering and brake | |
|---|---|
| 0x05 | Move steering left, right, or stop. Enforces steering limits and stall protection. |
| 0x06 | Move brake actuator forward, reverse, or stop. |
| 0x12 | Admin steering control, bypassing angle limits but retaining stall protection. |
| 0x13 | Admin brake control, bypassing normal control restrictions. |
| 0x14 | Automatically move steering to the centre of the configured steering range. |

| Vehicle controls | |
|---|---|
| 0x07 | Select speed mode 1, 2, or 3. |
| 0x08 | Select Drive, Neutral, or Reverse. |
| 0x09 | Turn off or select one light at a time. |
| 0x0A | Turn indicators off, left, right, or parking lights on. |
| 0x0B | Sound the horn for the selected duration. |

| System and configuration | |
|---|---|
| 0x0C | Reset the module and restore outputs to their defaults. |
| 0x0D | Save Wi-Fi SSID and password. |
| 0x0E | Set Wi-Fi to Off, Station, or Access Point mode. |
| 0x0F | Configure or read brake angle limits. |
| 0x10 | Configure or read steering angle limits. |
| 0x11 | Enable or disable steering limit enforcement. |
| 0x16 | Read the current state of all supported switches and controls. |

# 4. Command Examples

Bytes to transmit for each action. Append \n to every line.

| Action | Command | Action | Command |
|---|---|---|---|
| Get firmware version | `0x01` | Set gear to Reverse | `0x08 03` |
| Set acceleration to 50% | `0x02 32` | Turn on high beam | `0x09 02` |
| Read brake feedback | `0x03` | Turn on left indicator | `0x0A 01` |
| Read steering angle | `0x04` | Turn on parking lights | `0x0A 03` |
| Steer left | `0x05 01` | Sound horn for 0.5s | `0x0B 01` |
| Steer right | `0x05 02` | Reset module | `0x0C` |
| Stop steering | `0x05 00` | Read steering limits | `0x10` |
| Apply brake forward | `0x06 01` | Read steering limit status | `0x11` |
| Release brake | `0x06 00` | Centre steering | `0x14` |
| Set speed mode 2 | `0x07 02` | Read all statuses | `0x16` |

**Centre steering is asynchronous**
Command 0x14 starts an asynchronous centring move. The module may return a completion or error response **later** on the serial port - do not assume the reply is immediate, and always drain the response buffer before sending the next command.

## Sample session

Typical command sequence after connecting to the port (each line terminated with \n):

```
# connect at 115200 8N1
0x01 # firmware version 4.1.3
0x16 # read all switch states
0x08 01 # gear = Drive
0x07 02 # speed mode 2
0x02 32 # acceleration = 50%
0x05 01 # steer left
0x05 00 # stop steering
0x06 01 # brake on - accel now locked at 0%
0x02 32 # rejected: brake active
0x06 00 # brake released
0x14 # centre steering (response arrives later)
0x04 # read steering angle (0.1 deg units)
```

# 5. Important Restrictions

**BRAKE SAFETY**
Applying the brake automatically sets acceleration to 0%. Acceleration commands (0x02) are **rejected while the brake is active**. Release the brake (0x06 00) before requesting throttle.

**STEERING SAFETY**
Normal steering (0x05) is protected by configured angle limits and stall detection. Motion is stopped automatically when a limit is reached or the actuator stalls.

**ADMIN MODE**
Admin steering (0x12) bypasses angle limits, but **stall protection remains active**. Admin brake (0x13) bypasses normal control restrictions. Use only for service/recovery procedures.

**LOCKED COMMANDS (default)**
Wi-Fi configuration (`0x0D`, `0x0E`), steering limit enable/disable (`0x11`) and the longer horn pulses (`0x0B 02`, `0x0B 03`) are **locked by default** and rejected until unlocked.

**BRAKE LIMITS**
Brake angle limits (0x0F) can be configured and read back, but they are **not currently enforced** by the firmware.

**CENTRE STEERING**
The centring command (0x14) runs asynchronously and can return a completion or error response later.

## Quick safety matrix

| Command | Limits | Stall protect | Notes |
|---|---|---|---|
| 0x05 STR_SW | Enforced | Active | Normal operation |
| 0x12 STR_ADM | Bypassed | Active | Service / recovery |
| 0x06 BRK_SW | n/a | Active | Sets accel to 0% |
| 0x13 BRK_ADM | Bypassed | Active | Bypasses restrictions |
| 0x14 STR_CTR | Enforced | Active | Async response |
| 0x02 ACCEL | 20-90% | - | Rejected while braking |

**END OF DOCUMENT · ARGUS DRIVE SERIAL COMMAND INTERFACE · FIRMWARE v4.1.3**